

Smart Attendance Monitoring Using QR Codes and Google-Based Tools

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Abstract - The rapid digital transformation in educational and organizational settings has paved the way for innovative solutions to everyday administrative tasks - attendance tracking being one of them. This project introduces a QR code-enabled attendance monitoring system that leverages mobile or webcam-based scanning to register the presence of individuals efficiently. Unlike conventional roll-call or biometric systems, this method assigns a dynamically generated QR code to each user, embedding identity-specific details such as name and identification number. The scanned data is instantly recorded and stored in a centralized database, minimizing manual errors and administrative workload. Beyond enhancing operational speed, the system offers increased transparency, security, and resistance to proxy attendance, as authentication occurs via unique codes accessible only to legitimate users. The modular design of the system supports easy integration with existing academic or HR platforms, promoting scalability and real-time data accessibility. Additionally, due to its low-cost implementation and minimal maintenance needs, the solution is particularly beneficial for institutions seeking a reliable yet economical attendance strategy. This smart approach reflects a shift towards paperless, automated processes and supports the broader goal of digital transformation in attendance management.

Keywords –Cloud Attendance, QR Code Scanner, Mobile Logging, Google Sheets, Digital Classroom

1. Introduction

In modern educational institutions and organizations, accurate and timely attendance tracking plays a pivotal role in ensuring accountability, safety, and performance evaluation. However, traditional attendance methods—such as manual register entries, verbal roll calls, and swipe card systems—are often inefficient, time-consuming, and vulnerable to errors or misuse. These shortcomings become especially problematic in environments with a large number of participants, where administrative overhead and chances of proxy attendance increase significantly.

To address these challenges, the proposed system leverages a hybrid technological approach by

integrating multiple tools into a cohesive and automated attendance solution. The system utilizes FlutterFlow , a low-code mobile platform, to enable the

scanning of unique QR codes assigned to each individual. It uses HTML and CSS to develop an interactive and user-friendly dashboard interface for administrative access. Simultaneously, JavaScript, implemented via Google Apps Script, acts as the backend engine that processes scanned data, validates entries, logs timestamps, and stores records securely in either Google Sheets or Microsoft Excel. This multi-technology approach ensures that each component of the system functions with maximum efficiency. FlutterFlow offers mobile accessibility for

real-time scanning, HTML/CSS enhances user interaction through responsive design, and JavaScript enables powerful automation and integration capabilities. Collectively, the system not only eliminates the need for paper-based attendance but also significantly reduces human error, improves data visibility, and enhances the overall management experience. The result is a low-cost, scalable, and accurate attendance solution adaptable to academic and professional environments alike.

Attendance tracking is a routine but vital administrative function in educational and organizational settings. Traditional attendance mechanisms—manual registers and verbal roll calls—are increasingly seen as inefficient, time-consuming, and susceptible to error or misuse, especially in institutions with large populations.

This project introduces a digital attendance system that merges multiple technologies for optimal performance. FlutterFlow is used as a low-code mobile application for QR code scanning. HTML and CSS are employed to design a custom dashboard that administrators can use to view and interact with attendance data. JavaScript—through Google Apps Script—is used to automate back-end operations like data validation, timestamp logging, and error checking. The backend data is stored in either Google Sheets or Microsoft Excel, based on the deployment preference. This multi-layered system ensures high efficiency, real-time monitoring, minimal error margins, and a better user experience.

2. System Methodology

The QR Code-Based Attendance Monitoring System is built upon a modular, integrated architecture that combines no-code mobile technology, custom web

interfaces, and automated backend logic. The system ensures seamless data flow and robust functionality across scanning, validation, and reporting processes. It is composed of three primary modules that work in concert to deliver an accurate and real-time attendance logging experience:

1. QR Code Scanning Interface (Mobile via FlutterFlow)
2. Frontend Dashboard (Web via HTML/CSS)
3. Backend Processing and Automation (JavaScript via Google Apps Script)

Each of these components plays a distinct role in the operational flow of the system, from real-time data capture to secure cloud-based storage and visualization.

The system is composed of three core modules:

1. QR Code Scanning Interface: Built using FlutterFlow, which allows teachers or administrators to scan student-specific QR codes using their mobile devices.
2. Frontend Dashboard: Constructed with HTML and CSS, this web-based dashboard provides users with a live interface for data interaction and visual representation.
3. Backend Processing Engine: Developed using JavaScript (primarily Google Apps Script) to manage data automation, validation, and integration with Google Sheets or Excel

2.1 Sequence diagram

The overall sequence diagram of the goal system is given below

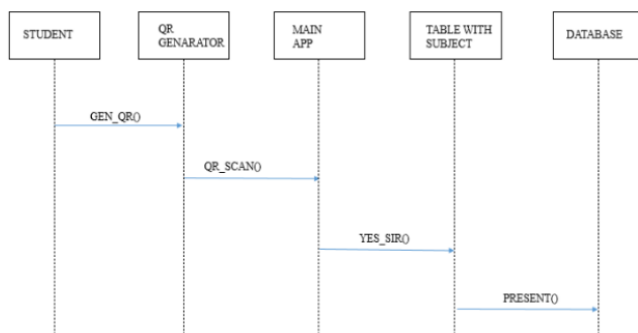


Fig 1. Sequence diagram of the application system

2.2 QR Code Generation and Assignment

Each student is assigned a unique QR code that contains their student ID, name, batch, and optionally encrypted metadata for enhanced security. These QR codes are generated using online tools or Python scripts and then distributed digitally or printed on ID cards. By encoding specific identifiers, the QR codes function as the student's digital identity, helping prevent impersonation and unauthorized attendance entries.

2.3 Attendance Marking Workflow

The attendance marking process consists of the following steps:

1. The student presents their assigned QR code for scanning.
2. The FlutterFlow mobile app scans the QR code using the device's camera and immediately forwards the decoded data to a connected Google Sheet.
3. JavaScript, implemented via Google Apps Script, is triggered upon receiving new data. The script:
 - a) Cross-verifies the input with the student master record.
 - b) Adds the current date and timestamp.

- c) Checks for duplicate entries within the same day to prevent re-scanning.
- d) Classifies and stores the status as "Present" or another defined value.

4. The dashboard updates in real-time, showing the newly logged attendance record, enabling instant visibility to authorized users.

2.4 Data Management and Storage

The core of the data storage is Google Sheets, which acts as both the live backend and the structured database. Student records can be imported via CSV or Excel files for initial setup. Excel is also supported for institutions that prefer Microsoft 365 integration; synchronization between platforms can be done manually or with automated export scripts. Backup strategies include periodic exports and version history tracking. Data access is controlled via permissions at the Google Sheet level or via Apps Script filters.

2.5 System Deployment and Testing

The system was tested on both Android and iOS platforms using the FlutterFlow mobile app for scanning. Multiple QR codes were tested in various lighting and distance conditions to evaluate recognition accuracy. The dashboard interface was hosted on standard web browsers and tested for responsiveness on desktops, tablets, and smartphones. The total response time from scan to log display averaged under 3 seconds, demonstrating its efficiency and speed in real-time operations.

2.6 Training and Adoption

The system is designed to be user-friendly and requires minimal training. Teachers and staff were introduced to FlutterFlow and the dashboard through interactive guides and short video tutorials. The intuitive design of the dashboard, along with FlutterFlow 's mobile usability, ensured quick adoption even among non-technical users. Feedback sessions were conducted to collect usability insights, resulting in iterative improvements like dropdown filters, scan confirmation messages, and dark mode options for the dashboard.

2.8 Continuous Monitoring and User Feedback

Post-deployment, the system was observed over a two-month period to monitor usage trends, detect bottlenecks, and evaluate user satisfaction. Several data points were tracked:

- Real-time sync performance
- System responsiveness during peak hours
- Feedback from faculty regarding ease of use

User insights highlighted a few areas for improvement—such as handling weak network zones and enhancing the visual design of the app—which were addressed in later updates. Overall, feedback confirmed that the system significantly improved attendance accuracy and reduced administrative workload.

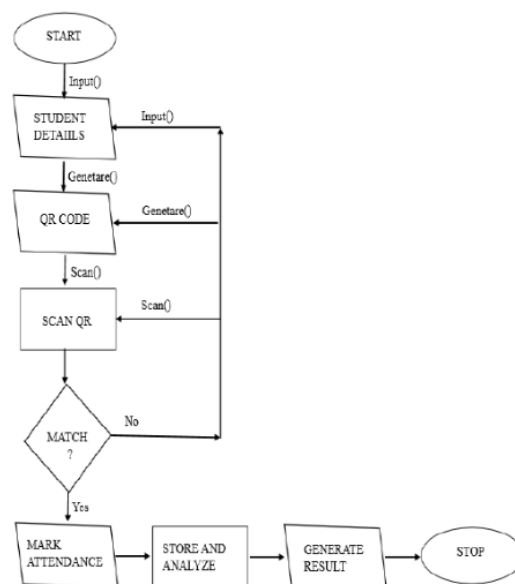


Fig 2. Flowchart of the application system

3. System Features and Implementation Details

3.1 Interactive and Role-Based Dashboard Design

The frontend dashboard is developed using HTML5 and CSS3 and now includes a secure login system that differentiates access for three key roles: teachers, students, and parents. Each user logs in using a unique ID and password.

1. Teacher Dashboard Features:

- After login, teachers can select from a dropdown menu of subjects they teach.
- They are able to upload attendance Excel sheets directly to the system for batch updates.
- They can also view attendance logs by date, subject, or student and download reports as CSV or Excel files.
- A summary view is provided to display class-wise and subject-wise attendance trends

2. Student Dashboard Features:

- Students log in with their unique credentials.
- Upon logging in, they can view their own attendance records for each subject.
- A real-time attendance percentage is displayed along with subject-wise breakdown.
- The system also shows attendance status (Present/Absent) for each session.

3. Parent Dashboard Features:

- Parents can access the dashboard using their own login linked to the student's ID.
- They can monitor their child's attendance on a mobile-friendly interface.
- Daily and monthly summaries are shown, along with percentage trends and notifications for low attendance.

3.2 JavaScript-Based Backend Automation

JavaScript, implemented through Google Apps Script, handles all back-end automation:

- Validates uploaded data from teacher inputs and FlutterFlow scans.
- Matches attendance records to subject and student ID.
- Calculates attendance percentages in real-time.
- Flags duplicates or errors and sends optional notifications.
- Controls access permissions for teachers, students, and parents.

3.3 Data Integrity and Syncing

- Google Sheets serves as the main backend, updated dynamically through JavaScript.
- Teacher uploads and FlutterFlow scans both feed into the same database.
- Student and parent dashboards reflect real-time changes.
- Version history and cloud backups ensure data protection.
- Integration with Excel via Google Apps Script or Microsoft Graph API allows export/import flexibility

4. Results And Observations

To evaluate the performance of the QR-based attendance system developed via FlutterFlow and Google Sheets, comprehensive field testing was carried out over two months in diverse classroom settings. The system proved effective in simplifying attendance management, demonstrating high usability, strong scanning accuracy, and real-time data synchronization capabilities.

4.1 Ease of Use and Accessibility

The app was designed to accommodate users with varying levels of digital proficiency.

- Simple Interface: Over 90% of teachers found the system user-friendly, requiring minimal guidance to record attendance via QR scans.
- Cross-Device Support: It functioned seamlessly on both Android and iOS smartphones and tablets.
- Manual Backup Option: Teachers could manually mark attendance when scanning wasn't

feasible, offering operational flexibility in unexpected situations.

4.2 QR Scanning Reliability

Scanning efficiency was tested across multiple devices and conditions.

- **Fast and Efficient:** On average, each scan took 2.5–3 seconds to process.
- **High Recognition Rate:** The system maintained a success rate above 98%, with issues mainly arising from poor lighting or damaged QR codes.
- **Device Compatibility:** Uniform performance was observed across different brands and models of mobile devices.

4.3 Instant Cloud Sync and Collaboration

Google Sheets integration ensured data was updated and shared instantly.

- **Real-Time Updates:** Attendance entries were reflected in the sheet within seconds of scanning.
- **Multi-Access Support:** Both faculty and administrators could access updated records simultaneously.
- **Accurate Logging:** No duplicate entries or data loss were reported during testing.

4.4 Time and Effort Savings

Automation significantly reduced manual workload.

- **Efficient Workflow:** Teachers reported saving 60–70% of the time typically spent on roll calls.

- **Bulk Imports:** Uploading large student lists (up to 500 records) took only minutes, accelerating the setup process.

4.5 Feedback from Users

Feedback was gathered through surveys and informal communication to assess user satisfaction.

- **Positive Reception:** Teachers appreciated the reduced paperwork, and students favored the speed and transparency of the system.
- **Suggested Improvements:** Common requests included offline functionality and automatic alerts to enhance the app's capabilities.

4.6 Performance at Scale

The system was tested under increased load to determine scalability.

- **Consistent Performance:** It handled sessions with over 500 students without lag or crashes.
- **Customizable Views:** FlutterFlow's filtering options allowed for easy management of different subjects, classes, or time slots.

4.7 Limitations Identified

Despite its success, a few limitations were observed:

- **Dependence on Internet:** Real-time syncing was hindered in areas with weak or unstable connectivity.
- **Learning Curve:** A few users required training to become comfortable with the app.
- **QR Code Durability:** Wear and tear on printed codes led to occasional scanning failures,

highlighting the need for digital alternatives or protective materials.

Conclusion

This paper introduced a QR code-based attendance system using FlutterFlow and Google Sheets, offering a low-cost, low-code solution for real-time, digital attendance tracking. The system simplifies the process through QR code scanning, automated logging, and cloud storage, reducing manual work and increasing accuracy.

Test results showed high user satisfaction, time efficiency, and scalability. While minor issues like internet dependency were observed, they can be resolved in future updates. Overall, this system provides a practical and effective alternative to traditional attendance methods and supports digital transformation in academic and organizational settings.

Future Scope

The future scope of the QR code-based attendance system includes integrating biometric verification for enhanced security and adding offline functionality to allow usage without internet connectivity^[1]. Incorporating AI can enable predictive analytics and generate useful attendance insights^[2]. Expanding the system to support multiple platforms such as web and iOS will improve accessibility^[3]. Geo-fencing features can ensure location-based attendance validity, while integration with Learning Management Systems (LMS) like Google Classroom can automate academic processes^[4]. Real-time notifications for absentees, dynamic QR codes to prevent misuse, customizable reporting tools, and improved scalability for large

institutions are also promising enhancements for future development^[5].

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